INCH-POUND
MIL-C-24640/23C
26 June 1995
SUPERSEDING
MIL-C-24640/23B

MILITARY SPECIFICATION SHEET

CABLE, ELECTRIC, 600 VOLTS, LIGHT WEIGHT, MULTI-CONDUCTOR, AWG 18, TYPE MXCW (INCLUDING VARIATION TYPES MXCOW AND MXCWA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation: MIL-C-24640.

CONSTRUCTION (Watertight)

FIRST - Copper conductor, uncoated AWG 18, 7 strands, in accordance with ASTM B-8, Nominal strand diameter 0.049 inch

SECOND - Mica glass tape insulation 0.0029 ±.0002 inch (50 percent overlap maximum and minimum overlap of 45 percent).

THIRD - Opaque-pigmented polyimide tape .001 inch (50 percent overlap minimum).

FOURTH - Opaque-pigmented polyimide tape .001 inch with 0.0001 inch Fluorinated Ethylene Propylene [FEP] (50 percent overlap minimum).

FIFTH - Modified aromatic polyimide resin coating 0.0005 inch minimum (.001 inch nominal). Identification codes applied using Method 1 or 3.

SIXTH - The required number of component wires shall be cabled together. Cabling sequence shall be consecutive, starting with number 1 from center outward. Fillers may be used as necessary, to form a round cable.

SEVENTH - Binder tape applied helically with overlap.

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CONSTRUCTION VARIANT FOR TYPE MXCW

EIGHTH - Crosslinked polyolefin jacket. (See Table I for thickness).

CONSTRUCTION VARIANT FOR TYPE MXCWA

EIGHTH - Crosslinked polyolefin jacket. (See Table II for thickness).

NINTH - Braided metal armor. Dimensions in accordance with Table II.

CONSTRUCTION VARIANT FOR TYPE MXCOW

EIGHTH - Braided shield, tin coated copper strands. Braid must be constructed to meet Surface Transfer Impedance requirements as listed herein.

NINTH - Binder tape applied helically with overlap.

TENTH - Crosslinked polyolefin jacket. (See Table IV for thickness).

* TABLE I. Construction details - unarmored.

PIN M24640/23	Type number	Number of conductors	Jacket thickness (inch) (minimum)	Cable diameter (inch) (min)	(max)	Cable weight (pounds per 1000 feet) (max)
-01UN	MXCW-7	7	0.038	0.295	0.319	90.0
-02UN	MXCW-10	10	.041	.375	.405	139.0
-03UN	MXCW-14	14	.041	.402	.434	173.0
-04UN	MXCW-19	19	.041	.440	.474	215.0
-05UN	MXCW-24	24	.044	.520	.560	295.0
-06UN	MXCW-30	30	.044	.547	.589	340.0
-07UN	MXCW-37	37	.044	.584	.630	398.0
-08UN	MXCW-44	44	.044	.656	.708	492.0
-09UN	MXCW-61	61	.047	.729	.785	640.0

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TABLE II. Construction details - armored.

PIN M24640/23	Type number	Number of conductors	Jacket thickness (inch) (minimum)	Cable diameter (inch) (min)	(max)	Cable weight (pounds per 1000 feet) (max)
-01AN	MXCWA-7	7	0.038	0.345	0.369	110.0
-02AN	MXCWA-10	10	.041	.425	.455	177.2
-03AN	MXCWA-14	14	.041	.452	.485	215.8
-04AN	MXCWA-19	19	.041	.490	.524	300.0
-05AN	MXCWA-24	24	.044	.570	.610	337.0
-06AN	MXCWA-30	30	.044	.597	.639	385.3
-07AN	MXCWA-37	37	.044	.634	.680	452.9
-08AN	MXCWA-44	44	.044	.706	.758	547.7
-09AN	MXCWA-61	61	.047	.779	.835	690.0

TABLE III. Construction details - overall shield.

PIN M24640/23	Type number	Number of conductors	Diameter over shield (inch) (nom)	Jacket thickness (inch) (minimum)	Cable diameter (inch) (min)	(max)	Cable weight (pounds per 1000 feet) (max)
-01UO	MXCOW-7	7	0.226	0.038	0.340	0.366	140.0
-02UO	MXCOW-10	10	.297	.041	.415	.447	185.0
-03UO	MXCOW-14	14	.325	.041	.440	.474	225.0
-04UO	MXCOW-19	19	.370	.041	.477	.515	280.0
-05UO	MXCOW-24	24	.441	.044	.557	.601	362.0
-06UO	MXCOW-30	30	.469	.044	.584	.630	399.0
-07UO	MXCOW-37	37	.514	.044	.622	.670	466.0
-08UO	MXCOW-44	44	.585	.047	.697	.751	574.0
-09UO	MXCOW-61	61	.652	.047	.757	.817	715.0

REQUIREMENTS:

Qualification required.
Construction details: See tables I, II and III.

Finished wire:

For polyimide tape constructions, the following additional tests shall be applicable in accordance with MIL-W-81381:

Temperature rating: 150°C (302°F) maximum conductor temperature
Voltage rating: 600 volts (rms) at sea level.
Blocking: Oven temperature, 200± 2°C (392± 3.6°F) no sticking of conductors

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Flammability: 3 seconds (max) after flame.
3.0 inches (max) flame travel.
No flaming of tissue paper.
Humidity resistance: 5 megohms for 1000 feet, minimum insulation resistance after humidity exposure.
Impulse dielectric test: 100 percent test. Impulse voltage at 8.0 kV.
Insulation resistance: 2500 megohms for 1000 feet (min).
Polyimide cure test: No cracking.
Shrinkage: 0.062 inch maximum at 180°C
Surface resistance: 5 megohms inches (min), initial and final readings.
Thermal shock: Oven temperature, 150± 2°C (302± 3.6°F). Maximum change in measurement: 0.031 inch.
Wet dielectric test: 2500 volts (rms).

EXAMINATION AND TESTS:

Basic electrical:	Requirements
Conductor Resistance - ohms/1000 feet at 25°C, maximum	7.07
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor	2000
Conductor to shield (MXCOW only)	1000
Shield to shield	Not applicable
Insulation resistance - megohms/1000 feet, minimum	
Conductor to conductor	2500
Conductor to overall shield (MXCOW only)	2500
Shield to shield (component shield only)	Not applicable
Shield to shield (component shield to overall shield	Not applicable
Conductor and shield continuity	No failure
Jacket flaws (type MXCOW only)	No failure
Group A:	
Visual and dimensional	No failure
Capacitance:	
Mutual capacitance - at 1 MHz, picofarads/foot (pF/ft), maximum	Not applicable
Capacitance unbalance - percent, maximum	Not applicable
Characteristic impedance - at 1 MHz, ohms	Not applicable
Watertightness (see table XI of MIL-C-24640 for limits of water leakage)	No failure
Weight	No failure
Group B:	
Cold bending	No failure
Drip	No failure
* Cross link proof test (percent maximum)	
Jacket	50
Tear - pounds per inch thickness, minimum	35

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	Physicals (unaged)	
	Jacket	
	Tensile strength - lb/in^2, minimum	1300
	Elongation - percent, minimum	160
	Attenuation - at 3 MHz, dB/100 feet, maximum	Not applicable
	Gas flame (1 hour)	No failure
	Group C:	
	Physicals (aged)	
	Jacket	
	Air oven	
	Tensile strength - percent of unaged, minimum	60
	Elongation - percent of unaged, minimum	60
	Hot oil immersion	
*	Tensile strength - percent of unaged, minimum	50
*	Elongation - percent of unaged minimum	50
*	Permanence of printing (conductor) - method 1 only cycles, minimum	25
*	Permanence of printing (jacket) - cycles, minimum	125
	Armor - conformance to material, (type MXCWA only) construction and coverage	No failure
	Shield - conformance to material, construction and coverage (type MXCOW only)	No failure
	Heat distortion - percent of unaged, maximum	30
*	Cable sealant removability	No failure
	Shrinkage	No failure
	Mutual inductance - dB/1000 feet, maximum	Not applicable
*	Surface transfer impedance (type MXCOW only)	
	Milliohms per meter, maximum	700
	EMP response, dB, minimum	60
	Pulse response time - ms/500 feet	
	Rise time, maximum	Not applicable
	Decay time, maximum	Not applicable
*		
	Group D	
	Flame propagation	No Failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

Aging and compatibility (cable)	No failure
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50

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*	Smoke index, maximum	
	Jacket	25
	Fillers	45
	Insulation	10
	Toxicity index, maximum	
	Jacket	5
	Fillers	5
	Insulation	1.5
	Durometer (jacket) - (type A) hardness, minimum	80
	Weathering (jacket)	No failure
	Fungus Resistance	No failure
	Abrasion resistance	No failure
	Gas Flame (3 hour)	Not applicable
	Component wire	No failure

Unit ordering length: 7 - 19 500 feet nominal.
24 - 61 250 feet nominal

NOTE: The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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DLA - IS

Preparing Activity:
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Agent:
DLA-IS

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